
MSML 605

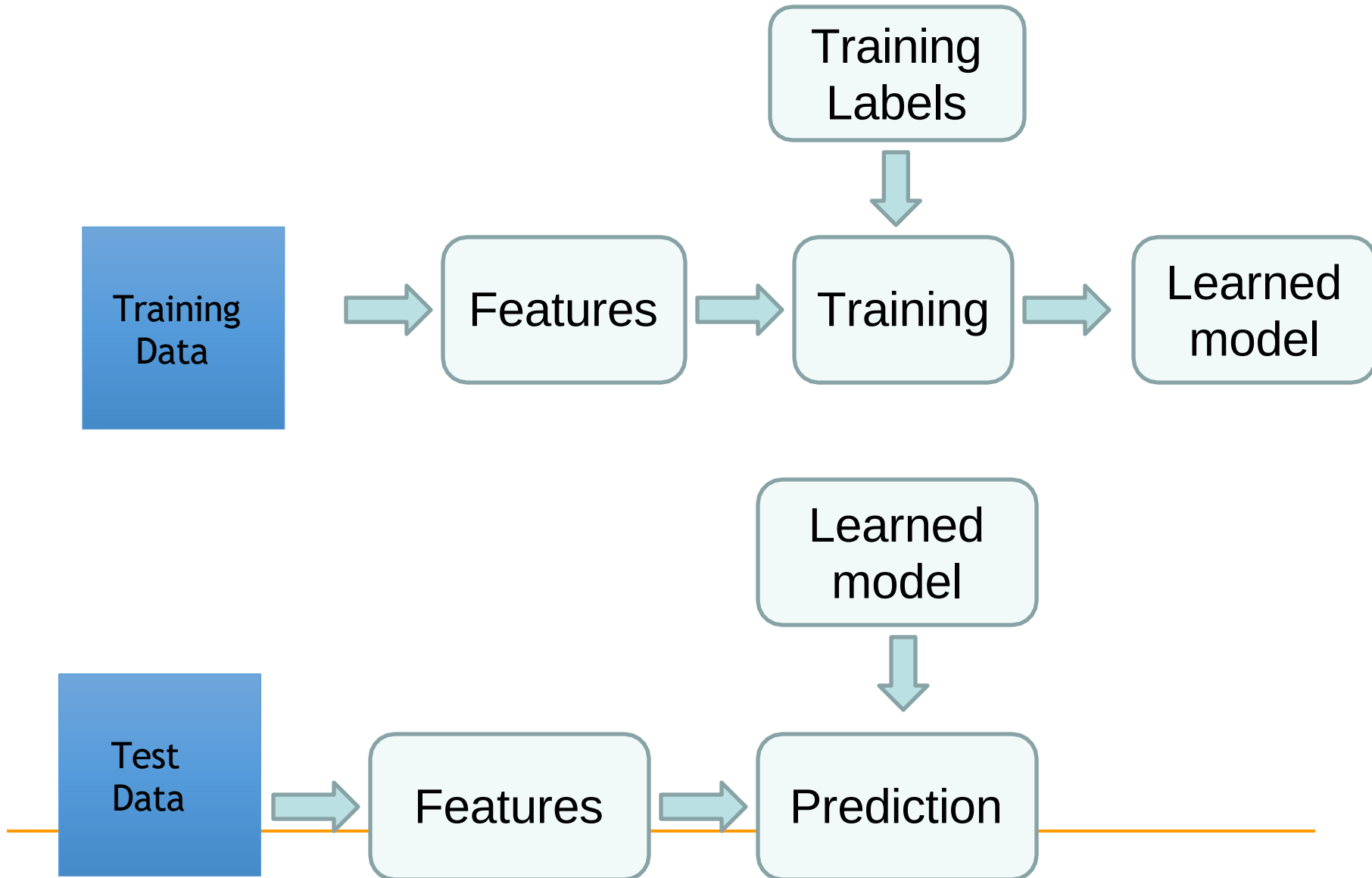
Computation

Machine Learning Hardware

- CPUs
- GPUs
- FPGAs
- Other accelerators



Machine Learning Project



Machine Learning hardware

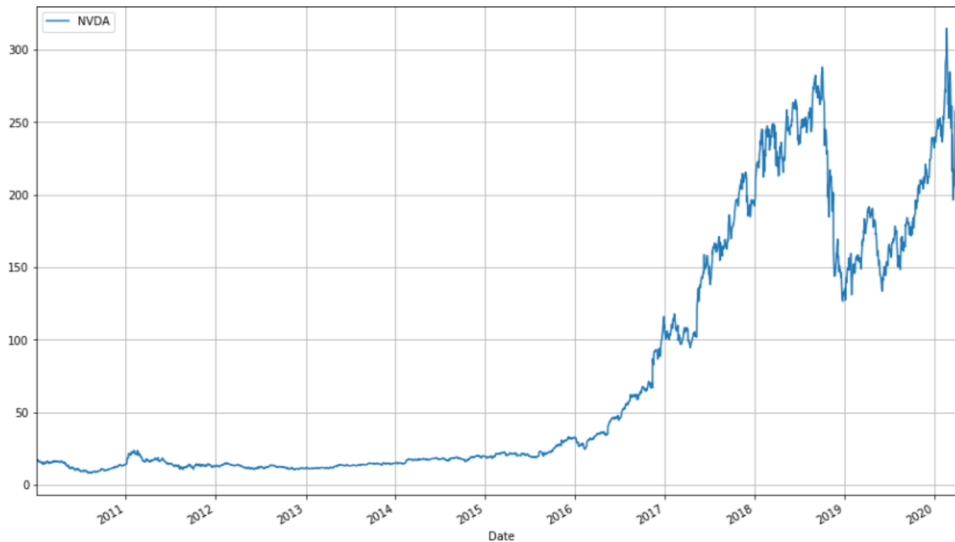
- *Speed up each block of the pipeline
for example, matrix-matrix multiplication,
convolution*
- *Data or memory paths for machine learning work
example: caching*
- *Application-specific functional units*

Processing

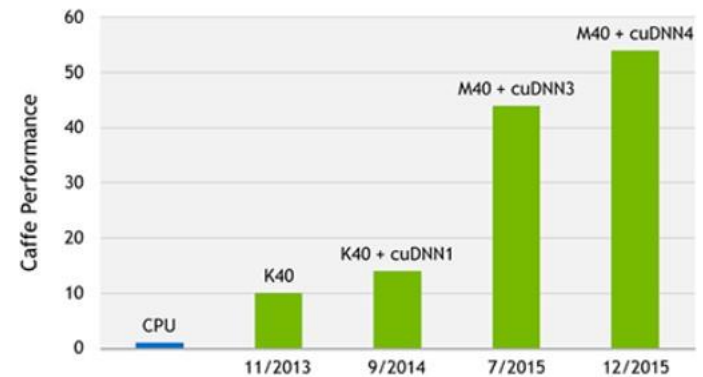
- *CPU is good at executing few complex operations.*
 - *In ML most of the processing involves matrix multiplication.*
 - *Lots of small calculations.*
 - *GPU is well suited for those kind of computations.*
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Processing

- *GPU utilizes parallel architecture.*
- *It is very good at handling many sets of very simple instructions.*

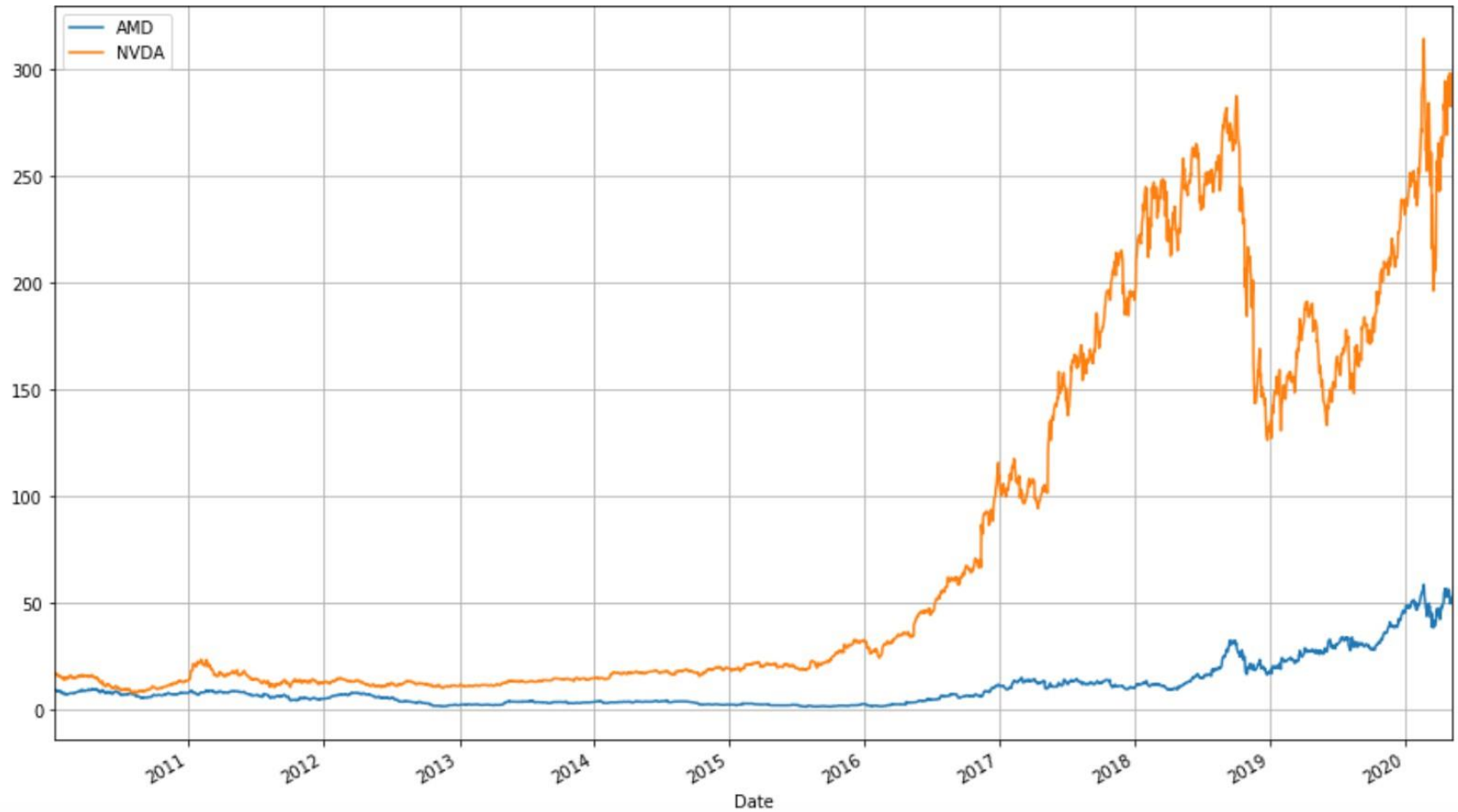


50X BOOST IN DEEP LEARNING IN 3 YEARS



AlexNet training throughput based on 20 iterations.
CPU: 1x E5-2680v3 12 Core 2.5GHz, 128GB System Memory, Ubuntu 14.04

GPU's



CPU vs. GPU



18 cores



2560 cores

CPU vs. GPU

	CPU i9 Xseries	GeForce GTX 1080
Cores	18 (36 threads)	2560
Clock Speed (GHz)	4.4	1.6G
Memory	Shared	8GB
Price (\$)	1799	549

GPU programming

- CUDA
 - C-like code that runs on GPU
 - Other APIs: cuBLAS, cuFFT, cuDNN, etc
- OpenCL
 - Similar to CUDA, but runs on CPU's as well
 - usually slower

Frameworks

- Caffe (Berkeley)
- Torch (NYU / Facebook)
- Theano (University of Montreal)

Frameworks

- Caffe (Berkeley)
- Caffe2 (Facebook)
- Torch (NYU / Facebook)
- PyTorch (Facebook)
- Theano (University of Montreal)
- TensorFlow (Google)
- Paddle (Baidu)
- CNTK (Microsoft)
- MXNet (Amazon)

DeepLearning Frameworks

- Computational graphs
- Gradient computation
- Run on GPU seamlessly